

Table 1: Test results for examples 1 – 3.

N	Pdim	SSDFP				SMDFP				PBDFFP				HHLSL				ILR				
		Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm	
1	5000	x ₀ ¹	1	2	0.0102	0	***	***	***	11	13	0.0593	7.12E-12	3	5	0.0276	0	23	31	0.0928	9.49E-11	
	5000	x ₀ ²	1	2	0.0049	0	11	24	0.0574	0	12	16	0.0550	3.91E-11	12	20	0.0462	4.84E-11	20	29	0.0707	3.00E-11
	5000	x ₀ ³	4	7	0.0068	2.34E-14	***	***	***	13	21	0.0644	1.92E-11	2	5	0.0115	0	23	33	0.0742	9.30E-11	
	5000	x ₀ ⁴	1	2	0.0029	0	***	***	***	22	26	0.0655	1.94E-11	11	18	0.0398	9.24E-11	20	27	0.0621	5.26E-11	
	5000	x ₀ ⁵	1	2	0.0031	0	***	***	***	11	13	0.0379	3.83E-11	2	4	0.0106	0	20	29	0.1857	1.56E-11	
	5000	x ₀ ⁶	1	2	0.0028	0	***	***	***	22	26	0.0719	1.94E-11	11	18	0.0403	8.89E-11	20	27	0.0653	5.27E-11	
	10000	x ₀ ¹	1	2	0.0036	0	***	***	***	11	13	0.0522	7.12E-12	3	5	0.0203	0	23	31	0.1046	9.49E-11	
	10000	x ₀ ²	1	2	0.0039	0	11	24	0.0672	1.17E-22	12	16	0.0664	5.53E-11	12	20	0.0602	6.85E-11	20	29	0.0950	4.24E-11
	10000	x ₀ ³	4	7	0.0163	3.32E-14	***	***	***	13	21	0.0728	2.71E-11	2	5	0.0157	0	23	34	0.1328	1.81E-11	
	10000	x ₀ ⁴	1	2	0.0053	0	***	***	***	22	26	0.1197	2.78E-11	12	18	0.0559	9.11E-13	20	27	0.0879	7.42E-11	
	10000	x ₀ ⁵	1	2	0.0060	0	***	***	***	11	13	0.0577	5.42E-11	2	4	0.0211	0	20	29	0.0914	2.21E-11	
	10000	x ₀ ⁶	1	2	0.0037	0	***	***	***	22	26	0.1030	2.79E-11	12	18	0.0574	8.85E-13	20	27	0.1112	7.41E-11	
	50000	x ₀ ¹	1	2	0.0113	0	***	***	***	11	13	0.1841	7.13E-12	3	5	0.0593	0	23	31	0.3804	9.49E-11	
	50000	x ₀ ²	1	2	0.0128	0	11	24	0.2398	1.03E-20	13	17	0.2252	6.99E-12	13	20	0.1998	2.40E-12	20	29	0.3509	9.49E-11
	50000	x ₀ ³	4	7	0.0408	7.41E-14	***	***	***	22	26	0.2472	6.06E-11	2	5	0.0519	0	23	34	0.3783	4.04E-11	
	50000	x ₀ ⁴	1	2	0.0115	0	***	***	***	22	26	0.3769	6.22E-11	12	18	0.1896	2.01E-12	26	33	0.4283	8.37E-11	
	50000	x ₀ ⁵	1	2	0.0123	0	***	***	***	12	14	0.2280	6.85E-12	2	4	0.0438	0	20	29	0.3808	4.93E-11	
	50000	x ₀ ⁶	1	2	0.0111	0	***	***	***	22	26	0.4290	6.22E-11	12	18	0.2039	2.00E-12	26	33	0.4283	8.37E-11	
2	5000	x ₀ ¹	9	11	0.0175	7.86E-11	***	***	***	11	13	0.0427	6.49E-12	18	25	0.0674	3.58E-11	***	***	***	***	
	5000	x ₀ ²	9	11	0.0231	5.18E-11	***	***	***	10	12	0.0501	7.54E-11	23	32	0.0864	4.38E-11	***	***	***	***	
	5000	x ₀ ³	9	10	0.0171	7.92E-11	13	26	0.0737	8.59E-11	10	12	0.0392	5.19E-11	20	28	0.0863	1.39E-11	***	***	***	***
	5000	x ₀ ⁴	9	11	0.0202	6.47E-11	***	***	***	10	12	0.0428	9.43E-11	22	30	0.0860	6.57E-11	***	***	***	***	
	5000	x ₀ ⁵	9	11	0.0185	6.46E-11	***	***	***	10	12	0.0458	9.41E-11	21	30	0.0758	2.35E-11	***	***	***	***	
	5000	x ₀ ⁶	9	11	0.0191	6.47E-11	***	***	***	10	12	0.0438	9.43E-11	22	30	0.1095	6.57E-11	***	***	***	***	
	10000	x ₀ ¹	10	11	0.0292	1.14E-11	***	***	***	11	13	0.0702	9.18E-12	18	24	0.1038	2.06E-11	***	***	***	***	
	10000	x ₀ ²	9	11	0.0249	7.31E-11	***	***	***	11	13	0.0884	6.05E-12	18	24	0.1109	5.10E-11	***	***	***	***	
	10000	x ₀ ³	9	11	0.0321	5.03E-11	***	***	***	10	12	0.0713	7.35E-11	17	23	0.0970	5.11E-11	***	***	***	***	
	10000	x ₀ ⁴	9	11	0.0295	9.14E-11	***	***	***	11	13	0.0789	7.56E-12	19	26	0.1365	4.44E-11	***	***	***	***	
	10000	x ₀ ⁵	9	11	0.0265	9.13E-11	***	***	***	11	13	0.0750	7.54E-12	21	29	0.1127	2.75E-11	***	***	***	***	
	10000	x ₀ ⁶	9	11	0.0249	9.14E-11	***	***	***	11	13	0.0725	7.55E-12	19	26	0.1112	4.43E-11	***	***	***	***	
	50000	x ₀ ¹	10	11	0.1179	2.54E-11	***	***	***	11	13	0.3261	2.05E-11	17	22	0.4207	4.77E-11	***	***	***	***	
	50000	x ₀ ²	10	11	0.1159	1.68E-11	***	***	***	11	13	0.3085	1.35E-11	21	29	0.5367	6.42E-11	***	***	***	***	
	50000	x ₀ ³	10	11	0.1127	1.15E-11	***	***	***	11	13	0.3069	9.37E-12	17	22	0.4303	3.01E-11	***	***	***	***	
	50000	x ₀ ⁴	10	11	0.1149	2.09E-11	***	***	***	11	13	0.3429	1.69E-11	21	28	0.5048	3.92E-11	***	***	***	***	
	50000	x ₀ ⁵	10	11	0.1204	2.09E-11	***	***	***	11	13	0.3604	1.69E-11	20	26	0.5232	9.60E-11	***	***	***	***	
	50000	x ₀ ⁶	10	11	0.2127	2.09E-11	***	***	***	11	13	0.3358	1.69E-11	21	28	0.5312	3.92E-11	***	***	***	***	
3	5000	x ₀ ¹	1	2	0.0054	0	***	***	***	10	13	0.0370	9.23E-11	3	5	0.0122	0	21	31	0.0615	1.69E-11	
	5000	x ₀ ²	1	2	0.0067	0	11	24	0.0512	0	13	17	0.0577	1.36E-11	24	37	0.0874	2.27E-11	28	45	0.0900	2.59E-12
	5000	x ₀ ³	4	7	0.0127	0	***	***	***	14	22	0.0766	1.10E-11	2	5	0.0150	0	28	45	0.0929	2.46E-12	
	5000	x ₀ ⁴	1	2	0.0057	0	***	***	***	11	14	0.0386	4.60E-11	25	39	0.0929	3.62E-11	24	35	0.0808	4.53E-12	
	5000	x ₀ ⁵	1	2	0.0065	0	***	***	***	12	14	0.0495	7.37E-12	2	4	0.0119	0	24	38	0.0773	4.93E-11	
	5000	x ₀ ⁶	1	2	0.0062	0	***	***	***	11	14	0.0387	4.61E-11	25	39	0.0983	3.62E-11	24	35	0.0797	4.40E-12	
	10000	x ₀ ¹	1	2	0.0058	0	***	***	***	10	13	0.0692	9.23E-11	3	5	0.0155	0	21	31	0.0985	1.69E-11	
	10000	x ₀ ²	1	2	0.0083	0	11	24	0.0612	1.17E-22	13	17	0.0657	1.93E-11	24	37	0.1007	3.21E-11	28	45	0.1392	3.66E-12
	10000	x ₀ ³	4	7	0.0185	0	***	***	***	14	22	0.0735	1.56E-11	2	5	0.0153	0	28	45	0.1623	3.48E-12	
	10000	x ₀ ⁴	1	2	0.0076	0	***	***	***	11	14	0.0636	6.51E-11	25	39	0.1068	5.11E-11	24	35	0.1007	6.37E-12	
	10000	x ₀ ⁵	1	2	0.0082	0	***	***	***	12	14	0.0582	1.04E-11	2	4	0.0114	0	24	38	0.1071	6.98E-11	
	10000	x ₀ ⁶	1	2	0.0083	0	***	***	***	11	14	0.0542	6.52E-11	25	39	0.1084	5.12E-11	24	35	0.1265	6.27E-12	
	50000	x ₀ ¹	1	2	0.0213	0	***	***	***	10	13	0.1914	9.23E-11	3	5	0.0496	0	21	31	0.3562	1.69E-11	
	50000	x ₀ ²	1	2	0.0230	0	***	***	***	13	17	0.2244	4.30E-11	24	37	0.3428	7.19E-11	28	45	0.5328	8.19E-12	
	50000	x ₀ ³	4	7	0.0700	0	***	***	***	14	22	0.3000	3.48E-11	2	5	0.0430	0	28	45	0.4773	7.79E-12	
	50000	x ₀ ⁴	1	2	0.0208	0	***	***	***	12	15	0.2062	8.24E-12	27	42	0.3725	2.03E-11	24	35	0.3977	1.41E-11	
	50000	x ₀ ⁵	1	2	0.0229	0	***	***	***	12	14	0.2167	2.33E-11	2	4	0.0412	0	28	44	0.4862	4.02E-12	
	50000	x ₀ ⁶	1	2	0.0221	0	***	***	***	12	15	0.2060	8.24E-12	27	42	0.3693	2.03E-11	24	35	0.4436	1.41E-11	

Table 2: Test results for examples 4 – 6.

N	Pdim	SSDFP				SMDFP				PBDFFP				HHLSL				ILR					
		Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm	Nit	Fval	Ptime	Norm		
4	5000	x_0^1	41	217	0.2050	6.61E-11	***	***	***	154	238	0.5648	9.83E-11	31	123	0.1269	7.58E-11	***	***	***	***		
	5000	x_0^2	40	217	0.2059	4.14E-11	***	***	***	161	246	0.6062	9.89E-11	30	121	0.1204	5.62E-11	***	***	***	***		
	5000	x_0^3	36	196	0.1769	7.45E-11	***	***	***	147	250	0.5775	9.98E-11	40	156	0.1667	6.38E-11	***	***	***	***		
	5000	x_0^4	47	246	0.2197	6.53E-11	***	***	***	128	221	0.4953	6.46E-11	25	103	0.1100	7.85E-11	***	***	***	***		
	5000	x_0^5	34	193	0.1757	7.09E-11	***	***	***	154	227	0.5743	9.65E-11	32	128	0.1671	5.01E-11	***	***	***	***		
	5000	x_0^6	41	220	0.2231	4.20E-11	***	***	***	158	237	0.6050	9.21E-11	30	120	0.1297	5.68E-11	***	***	***	***		
	10000	x_0^1	42	224	0.3143	8.19E-11	***	***	***	132	224	0.7964	7.96E-11	26	106	0.1608	9.86E-11	***	***	***	***		
	10000	x_0^2	35	191	0.2603	9.41E-11	***	***	***	159	252	0.8876	5.88E-11	26	106	0.1681	8.07E-11	***	***	***	***		
	10000	x_0^3	42	228	0.3067	6.61E-11	***	***	***	164	257	0.9272	9.45E-11	47	180	0.2728	6.90E-11	***	***	***	***		
	10000	x_0^4	40	211	0.2885	5.05E-11	***	***	***	134	208	0.7300	9.66E-11	34	134	0.2133	8.12E-11	***	***	***	***		
	10000	x_0^5	35	188	0.2633	8.97E-11	***	***	***	139	235	0.7854	8.58E-11	29	116	0.1856	5.55E-11	***	***	***	***		
	10000	x_0^6	32	180	0.2588	9.03E-11	***	***	***	164	234	0.8721	7.24E-11	27	110	0.1757	4.83E-11	***	***	***	***		
	50000	x_0^1	40	224	1.2320	6.39E-11	***	***	***	170	241	3.8722	8.15E-11	37	143	0.9183	9.53E-11	***	***	***	***		
	50000	x_0^2	36	196	1.0817	6.55E-11	***	***	***	172	258	4.0120	9.54E-11	27	109	0.7377	9.95E-11	***	***	***	***		
	50000	x_0^3	35	196	1.0725	2.76E-11	***	***	***	178	268	4.2874	9.17E-11	49	185	1.2389	9.97E-11	***	***	***	***		
	50000	x_0^4	41	218	1.2738	9.44E-11	***	***	***	144	228	3.5142	7.03E-11	31	123	0.7892	9.98E-11	***	***	***	***		
	50000	x_0^5	28	165	0.9171	6.82E-11	***	***	***	138	227	3.4511	9.56E-11	43	163	1.0612	9.61E-11	***	***	***	***		
	50000	x_0^6	40	218	1.3015	8.98E-11	***	***	***	161	242	3.7493	6.31E-11	27	110	0.7122	9.43E-11	***	***	***	***		
5	5000	x_0^1	7	23	0.0422	0	***	369	0.6402	1.12E-11	57	79	0.2639	3.94E-11	13	39	0.0645	9.87E-11	2	10	0.0194	0	
	5000	x_0^2	5	17	0.0284	0	***	30	0.5364	6.04E-11	69	98	0.3194	6.59E-11	2	7	0.0137	0	1	6	0.0150	0	
	5000	x_0^3	9	31	0.0488	2.38E-11	***	29	0.5742	1.77E-11	85	120	0.3953	7.66E-11	16	44	0.0789	0	4	16	0.0382	0	
	5000	x_0^4	1	3	0.0065	0	***	***	***	108	122	0.4542	9.40E-11	1	4	0.0097	0	1	6	0.0159	0		
	5000	x_0^5	1	3	0.0072	0	***	***	***	69	85	0.3044	8.33E-11	1	4	0.0084	0	1	6	0.0160	0		
	5000	x_0^6	1	3	0.0082	0	***	***	***	45	68	0.2207	8.39E-11	1	4	0.0078	0	1	6	0.0126	0		
	10000	x_0^1	7	23	0.0553	0	***	45	492	1.0941	6.99E-11	73	91	0.4367	8.80E-11	13	39	0.0854	9.92E-11	2	10	0.0295	0
	10000	x_0^2	5	17	0.0421	0	***	26	261	0.6176	2.67E-11	72	96	0.4486	8.60E-11	2	7	0.0182	0	1	6	0.0200	0
	10000	x_0^3	10	33	0.0760	0	***	36	340	0.8267	1.44E-11	70	107	0.4668	6.98E-11	9	25	0.0652	0	4	16	0.0481	0
	10000	x_0^4	1	3	0.0096	0	***	***	***	70	97	0.4490	5.98E-11	1	4	0.0133	0	1	6	0.0235	0		
	10000	x_0^5	1	3	0.0099	0	***	19	229	0.5072	9.75E-12	64	87	0.4101	7.22E-11	1	4	0.0125	0	1	6	0.0203	0
	10000	x_0^6	1	3	0.0104	0	***	***	***	51	69	0.3553	3.47E-11	1	4	0.0119	0	1	6	0.0196	0		
	50000	x_0^1	7	23	0.1831	0	***	45	494	3.6816	9.38E-11	71	89	1.5818	8.60E-11	13	39	0.3056	9.96E-11	2	10	0.0969	0
	50000	x_0^2	5	17	0.1412	0	***	23	234	1.9442	6.17E-11	113	138	2.3924	9.63E-11	2	7	0.0631	0	1	6	0.0648	0
	50000	x_0^3	8	28	0.2054	5.88E-13	***	29	260	2.2040	9.77E-12	98	127	2.1175	3.10E-11	15	41	0.3417	0	4	15	0.1688	0
	50000	x_0^4	1	3	0.0331	0	***	***	***	62	85	1.4508	3.29E-11	1	4	0.0347	0	1	6	0.0731	0		
	50000	x_0^5	1	3	0.0331	0	***	19	227	1.7746	6.53E-12	58	81	1.3262	7.42E-11	1	4	0.0322	0	1	6	0.0656	0
	50000	x_0^6	1	3	0.0320	0	***	***	***	39	55	0.9098	2.70E-11	1	4	0.0340	0	1	6	0.0683	0		
6	5000	x_0^1	5	22	0.0329	0	***	133	0.2280	9.88E-11	69	89	0.3035	9.05E-11	12	38	0.0611	0	1	7	0.0150	0	
	5000	x_0^2	3	13	0.0230	0	***	19	223	0.3769	5.56E-11	73	103	0.3547	8.75E-11	18	62	0.1196	5.18E-11	1	7	0.0167	0
	5000	x_0^3	5	15	0.0261	0	***	16	138	0.2510	9.86E-12	79	123	0.3906	7.98E-11	22	71	0.1363	2.06E-11	6	27	0.0563	0
	5000	x_0^4	1	4	0.0089	0	***	10	120	0.2022	3.16E-11	62	81	0.3172	9.67E-11	26	84	0.1576	2.56E-11	1	7	0.0168	0
	5000	x_0^5	1	4	0.0096	0	***	7	121	0.2441	8.32E-12	64	83	0.3448	9.00E-11	17	56	0.1098	9.57E-11	1	7	0.0154	0
	5000	x_0^6	1	4	0.0072	0	***	10	85	0.1465	2.91E-11	35	51	0.1876	8.74E-11	26	83	0.1548	4.40E-11	1	7	0.0155	0
	10000	x_0^1	5	22	0.0428	0	***	11	133	0.3136	9.88E-11	71	90	0.4261	9.45E-11	12	38	0.1098	0	1	7	0.0186	0
	10000	x_0^2	3	13	0.0323	0	***	12	143	0.3328	8.95E-11	89	114	0.5443	9.02E-11	19	62	0.1755	1.87E-11	1	7	0.0252	0
	10000	x_0^3	5	15	0.0434	0	***	14	134	0.3167	4.36E-11	96	128	0.6084	5.88E-11	21	68	0.1833	3.36E-11	5	24	0.0730	0
	10000	x_0^4	1	4	0.0107	0	***	10	120	0.2789	2.10E-11	53	75	0.3769	9.42E-11	24	77	0.2036	5.61E-11	1	7	0.0210	0
	10000	x_0^5	1	4	0.0117	0	***	10	121	0.2864	8.26E-12	59	80	0.3772	5.59E-11	17	55	0.1522	7.40E-11	1	7	0.0268	0
	10000	x_0^6	1	4	0.0123	0	***	7	85	0.1958	1.46E-11	30	46	0.1965	9.59E-11	25	80	0.2102	8.64E-11	1	7	0.0233	0
	50000	x_0^1	5	22	0.1466	0	***	11	133	1.1331	9.88E-11	83	100	1.7759	6.53E-11	12	38	0.3760	0	1	7	0.0763	0
	50000	x_0^2	3	13	0.0972	0	***	13	157	1.2880	2.06E-11	67	89	1.4470	9.31E-11	20	65	0.5541	7.01E-11	1	7	0.0774	0
	50000	x_0^3	5	15	0.1541	0	***	6	38	0.3877	0	85	125	1.9962	7.78E-11	20	67	0.6595	4.19E-11	5	22	0.2331	0
	50000	x_0^4	1	4	0.0365	0	***	10	120	0.9640	9.18E-12	54	77	1.3058	4.09E-11	25	80	0.8083	8.25E-11	1	7	0.0740	0
	50000	x_0^5	1	4	0.0330	0	***	10	121	0.9400	8.21E-12	61	79	1.3797	9.25E-11	17	55	0.5475	4.86E-11	1	7	0.0757	0
	50000	x_0^6	1	4	0.0368	0	***	6	73	0.6064	4.98E-11	19	34	0.5049	9.42E-11	26	83	0.8138	3.93E-11	1	7	0.0787	0